

The minimum fourth rank for the fixed Young-cubed prefix

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Abstract

Fix the entire Young-cubed poset through rank three. We prove that the minimum number of nonempty fourth-rank lower-cover blocks satisfying its point and pair incidence equations is exactly 50, allowing arbitrary repeated blocks. The upper bound is already supplied by the construction of Dai, Deng, Shi, Wu, and Yang. Our contribution is a matching lower bound: weights on the 22 rank-three elements and their pairs give an elementary inequality valid for every allowable block. We include an explicit 50-block witness and an exact verification supplement. The reflection extension shows that 50 is also the minimum fourth-rank size among infinite 3-differential posets retaining this complete prefix. The unrestricted minimum over all 3-differential posets is not determined.

1 The fixed-prefix problem

Let $\mathbf{Y}$ be Young's lattice of integer partitions, ordered by inclusion of their Ferrers diagrams. Set $R_s = (\mathbf{Y}^3)_s$: its elements are triples of partitions with total size s , and a cover adds one cell in one coordinate. We fix $R_0, \dots, R_3$ together with every cover relation. Their sizes are 1, 3, 9, 22. For $x, y \in R_3$ with $x \neq y$, write

$$d(x) = |\{u \in R_2 : u \triangleleft x\}|, \quad c(x, y) = |\{u \in R_2 : u \triangleleft x, u \triangleleft y\}|.$$

A feasible family is a finite indexed family $(B_z)_{z \in Z}$ of nonempty subsets of R_3 satisfying

$$\sum_{z \in Z} \mathbf{1}_{x \in B_z} = d(x) + 3 \quad (x \in R_3), \quad (1)$$

$$\sum_{z \in Z} \mathbf{1}_{\{x, y\} \subseteq B_z} = c(x, y) \quad (x, y \in R_3, x \neq y). \quad (2)$$

Different indices may specify identical blocks. Equation (2) includes every pair with $c(x, y) = 0$. Let N_* be the minimum of $|Z|$ over feasible families.

A 3-differential poset is a locally finite graded poset with a least element in which an element with d lower covers has $d + 3$ upper covers, and two distinct elements of the same rank have equally many common upper and common lower covers. Its ranks are finite: induction from the least element uses the finiteness of the number of upper covers at each element. The lower-cover sets of its rank-four elements satisfy (1)–(2) when its prefix is the one fixed above.

Theorem 1. *For the complete fixed prefix $(\mathbf{Y}^3)_{\leq 3}$, one has $N_* = 50$. Equivalently,*

$$\min\{|P_4| : P \text{ is an infinite 3-differential poset and } P_{\leq 3} \cong (\mathbf{Y}^3)_{\leq 3}\} = 50.$$

The isomorphism here preserves the whole order, including all covers.

Dai, Deng, Shi, Wu, and Yang [1, Sections 2.2–2.4] replace thirteen rank-four lower-cover blocks of $\mathbf{Y}^3$ by twelve, retaining the complete prefix and decreasing the fourth-rank size from 51 to 50. Their Proposition 3.1 gives the infinite extension. Theorem 1 supplies the matching lower bound for this same fixed prefix. An explicit feasible family is included below for self-contained verification; the upper bound itself is not claimed as a new construction. For adjacent use of block-incidence inequalities in extremal set systems, see Zeng [2]. That reference serves as methodological context; the proof here follows directly from (1)–(2).

2 An incidence-weight lower bound

Let $S \subset R_3$ consist of the six triples whose only nonempty coordinate is (3) or (1, 1, 1). Each $s \in S$ has exactly one lower cover, namely (2) or (1, 1) in that coordinate. These six lower covers are distinct. Hence $c(s, s') = 0$ for distinct $s, s' \in S$, so every feasible block meets S in at most one element.

The four shape types give the following counts.

Type in rank three	Number	$d(x)$	$d(x) + 3$
One coordinate (3) or (1, 1, 1)	6	1	4
One coordinate (2, 1)	3	2	5
One size-two coordinate and one (1)	12	2	5
Three coordinates (1)	1	3	6

In particular,

$$\sum_{s \in S} (d(s) + 3) = 24, \quad \sum_{x \notin S} (d(x) + 3) = 81. \quad (3)$$

At rank two, each of the six elements supported in one coordinate has four upper covers in R_3 : two enlarge that coordinate, and two add a cell in another coordinate. Each of the other three elements has five upper covers: four enlarge an occupied coordinate and one fills the empty coordinate. Double counting a pair once for each common lower cover therefore gives

$$\sum_{\{x, y\} \subset R_3} c(x, y) = 6 \binom{4}{2} + 3 \binom{5}{2} = 66. \quad (4)$$

Here and below the pair sums run over unordered two-element subsets. The unique lower cover of each $s \in S$ has three other upper covers. There are no contributing pairs inside S , whence

$$\sum_{\substack{\{x, y\} \subset R_3 \\ \{x, y\} \cap S \neq \emptyset}} c(x, y) = 6 \cdot 3 = 18. \quad (5)$$

Both counts retain common-lower-cover multiplicities.

Call a nonempty block *allowable* if every pair in it has positive $c(x, y)$. Every block in a feasible family is allowable. Give points and allowable pairs the weights

$$\alpha_x = \begin{cases} 1 & x \in S, \\ 2/3 & x \notin S, \end{cases} \quad \beta_{xy} = \begin{cases} -2/3 & \{x, y\} \cap S \neq \emptyset, \\ -1/3 & \{x, y\} \cap S = \emptyset. \end{cases}$$

Weights of zero pairs may be set to zero, since such pairs never occur in an allowable block. Define

$$\text{wt}(B) = \sum_{x \in B} \alpha_x + \sum_{\{x, y\} \subset B} \beta_{xy}.$$

Lemma 1. *Every nonempty allowable block B has $\text{wt}(B) \leq 1$.*

Proof. Put $k = |B|$ and $t = |B \cap S| \in \{0, 1\}$. The point weights sum to $(2k + t)/3$, and exactly $t(k - 1)$ of the pairs meet S . Thus

$$\text{wt}(B) = \frac{2k - \binom{k}{2} + t(2 - k)}{3}, \quad 1 - \text{wt}(B) = \frac{(k - 2)(k - 3 + 2t)}{6}. \quad (6)$$

For $k = 1$ the last expression is $(1 - t)/3$, and for $k = 2$ it is zero. For $k \geq 3$ both factors in the numerator are nonnegative. $\square$

Summing the lemma over a feasible indexed family and then applying its point and pair equations yields

$$\begin{aligned} |Z| &\geq \sum_{z \in Z} \text{wt}(B_z) \\ &= 24 + \frac{2}{3} \cdot 81 - \frac{1}{3} \cdot 66 - \frac{1}{3} \cdot 18 = 50. \end{aligned} \quad (7)$$

Every copy of a repeated block appears separately in this sum. Negative pair weights are legitimate because the constraints are equalities. The inequality covers blocks of every size and does not require a clique enumeration or a numerical optimization bound.

3 Attainment and infinite extension

Appendix A gives a family of 50 blocks: six singletons, 33 pairs, and 11 triples. Let D be the 22×9 zero-one matrix of lower covers listed there. Direct incidence counting gives

$$\sum_B \mathbf{1}_{i \in B} \mathbf{1}_{j \in B} = (DD^\top)_{ij} + 3\mathbf{1}_{i=j} \quad (0 \leq i \leq j \leq 21). \quad (8)$$

The displayed labels, lower sets, and blocks specify every entry of this identity; the supplement replays all 253 entries, including the 165 zero pairs, in integer arithmetic. This proves feasibility and hence $N_* \leq 50$. Together with (7), it proves the finite assertion of Theorem 1.

We give the reflection argument for completeness, following the construction in [1, Proposition 3.1]. Start with any feasible family. Attach one distinct rank-four element for each index z , with lower covers exactly B_z , and take the transitive closure together with the fixed prefix. Edges always raise rank by one, so this is a graded partial order and creates no extra adjacent-rank covers. Nonempty blocks give every new element a chain down to the least element. Equations (1)–(2) are exactly the two differential identities at rank three. The identities below rank three already hold in $\mathcal{Y}^3$.

Suppose now that a finite prefix through rank $n \geq 4$ satisfies these identities at every rank below n . For each $y \in P_{n-1}$, add a distinct element $y^* \in P_{n+1}$ whose lower covers are all the elements of P_n covering y . Also add three distinct elements with lower-cover block $\{x\}$ for each $x \in P_n$. The block of y^* is nonempty, since the identity at y gives $d(y) + 3 \geq 3$ upper covers. All new elements are distinct even when their blocks coincide.

An element $x \in P_n$ has one upper cover y^* for each of its lower covers and three further singleton upper covers. Two distinct elements of P_n have one common new upper cover for each common lower cover and none from the singleton blocks. The differential identities therefore hold at rank n , with all earlier relations preserved. The next rank is finite, of size $|P_{n-1}| + 3|P_n|$, and the nonempty blocks preserve the least element and gradedness.

Iteration produces nonempty finite ranks; each interval lies in finitely many such ranks and is finite. Each differential identity remains fixed after the next rank has been added. The union is consequently an infinite 3-differential poset retaining the given prefix through rank four. Conversely, any infinite 3-differential poset with the prescribed prefix provides a feasible family from its rank-four lower-cover sets. The two formulations have the same attainable fourth-rank sizes, proving the remaining assertion of Theorem 1.

4 Verification and scope

The supplement regenerates partition triples and their covers, checks the witness, and verifies the weights scaled by 3 using integers only. The positive-pair graph has 178 nonempty cliques: respectively 22, 66, 66, 21, 3 of sizes 1, 2, 3, 4, 5, and none of size six or larger. All cliques, including singletons and nonmaximal cliques, are admissible variables in the finite formulation; the 88 positive point/pair equations, with nonnegative integer multiplicities, encode the original family problem exactly. The clique condition enforces the zero pairs. The scaled weight sums are 3 on 120 cliques, 2 on 49, and 0 on 9; the scaled objective is 150. These finite checks provide a replay of the certificate. The lower-bound proof is the uniform inequality (6), independently of those counts.

The entire prefix is essential to the statement. Fixing just the four rank sizes would be a different problem. We make no claim about the unrestricted minimum $m_3(4)$, other differential parameters or ranks, or the classification of minimizers. The literature comparison is with the inspected version of [1]: it supplies the upper bound and the reflection construction, but no matching fixed-prefix lower bound. The bounded search recorded on 9 September 2026 did not find such a bound; it does not establish worldwide priority or an exhaustive account of the literature.

Computational and AI assistance. OpenAI Codex assisted with model formulation, certificate discovery, proof development, code, and manuscript preparation. Floating-point linear and mixed-integer optimization suggested the witness and weights; the reported numerical optimum is not used as proof. Separate exact implementations regenerated the finite model and checked both certificates. A separate model-context review checked the argument and independently replayed the data. These are computational and AI-assisted checks, not human peer review or proof-assistant formalization. The accompanying Python verifier uses only the standard library and writes its report to standard output.

A An explicit incidence certificate

Write 0 for the empty partition and concatenate parts, so $21 = (2, 1)$. The labels below fix the ordering used throughout the certificate; they differ from the numbering in [1]. The rank-two labels are

$$\begin{array}{lll} u_0 = (0; 0; 11) & u_1 = (0; 0; 2) & u_2 = (0; 1; 1) \\ u_3 = (0; 11; 0) & u_4 = (0; 2; 0) & u_5 = (1; 0; 1) \\ u_6 = (1; 1; 0) & u_7 = (11; 0; 0) & u_8 = (2; 0; 0) \end{array}$$

The lower covers of every rank-three element are listed in Table 1. A table entry u_j means $D_{ij} = 1$; all other entries are zero. In this numbering $S = \{0, 2, 6, 9, 17, 21\}$.

Table 1: Rank-three labels and their complete lower-cover sets.

i	Triple	Lower covers	$d + 3$	i	Triple	Lower covers	$d + 3$
0	(0; 0; 111)	u_0	4	11	(1; 0; 2)	u_1, u_5	5
1	(0; 0; 21)	u_0, u_1	5	12	(1; 1; 1)	u_2, u_5, u_6	6
2	(0; 0; 3)	u_1	4	13	(1; 11; 0)	u_3, u_6	5
3	(0; 1; 11)	u_0, u_2	5	14	(1; 2; 0)	u_4, u_6	5
4	(0; 1; 2)	u_1, u_2	5	15	(11; 0; 1)	u_5, u_7	5
5	(0; 11; 1)	u_2, u_3	5	16	(11; 1; 0)	u_6, u_7	5
6	(0; 111; 0)	u_3	4	17	(111; 0; 0)	u_7	4
7	(0; 2; 1)	u_2, u_4	5	18	(2; 0; 1)	u_5, u_8	5
8	(0; 21; 0)	u_3, u_4	5	19	(2; 1; 0)	u_6, u_8	5
9	(0; 3; 0)	u_4	4	20	(21; 0; 0)	u_7, u_8	5
10	(1; 0; 11)	u_0, u_5	5	21	(3; 0; 0)	u_8	4

The following blocks specify 50 distinct rank-four elements. Each block occurs once in this particular witness; this does not restrict allowed multiplicities in the optimization problem. Six singletons:

$$\{0\} \quad \{2\} \quad \{6\} \quad \{9\} \quad \{17\} \quad \{21\}$$

Thirty-three pairs:

$$\begin{array}{llllll}
\{0, 1\} & \{0, 3\} & \{0, 10\} & \{1, 2\} & \{1, 4\} & \{1, 11\} \\
\{2, 4\} & \{2, 11\} & \{3, 5\} & \{3, 7\} & \{4, 11\} & \{5, 6\} \\
\{5, 8\} & \{6, 8\} & \{6, 13\} & \{7, 8\} & \{7, 9\} & \{8, 9\} \\
\{9, 14\} & \{10, 15\} & \{10, 18\} & \{13, 16\} & \{13, 19\} & \{14, 16\} \\
\{14, 19\} & \{15, 17\} & \{15, 20\} & \{16, 17\} & \{17, 20\} & \{18, 20\} \\
\{18, 21\} & \{19, 21\} & \{20, 21\} & & &
\end{array}$$

Eleven triples:

$$\begin{array}{llll}
\{1, 3, 10\} & \{3, 4, 12\} & \{4, 5, 7\} & \{5, 12, 13\} \\
\{7, 12, 14\} & \{8, 13, 14\} & \{10, 11, 12\} & \{11, 15, 18\} \\
\{12, 15, 16\} & \{12, 18, 19\} & \{16, 19, 20\} &
\end{array}$$

All data in these tables are also in `certificate.json`. From the source directory, run

`python3 verify.py`

to regenerate the prefix, check (8), enumerate the allowable blocks, and verify the scaled weight inequalities. The same check tests the first reflection step, which adds $22 + 3 \cdot 50 = 172$ elements; the infinite conclusion rests on the proof above.

References

- [1] Xinan Dai, Wenhao Deng, Yingdong Shi, Tailin Wu, and Yuchen Yang, *An Explicit Counterexample to Stanley’s Rankwise Lower-Bound Conjecture for Differential Posets*, arXiv:2607.22988v2 (28 July 2026). <https://arxiv.org/abs/2607.22988v2>.
- [2] Zijian Zeng, *Uniform Block-Projection and Matching Bounds in a Frankl Three-Cover Branch*, 2026. <https://doi.org/10.2139/ssrn.7374798>.